# Networks and Complexity

## Solution 17-8

*This is an example solution from the forthcoming book Networks and Complexity. Find more exercises at https://github.com/NC-Book/NCB*

**Ex 17.8: Lotka-Volterra [Lvl. 3]**

The Lotka-Volterra model describes the interaction of a prey population $X$ and a predator population $Y$ by

$$\begin{aligned}\dot{X} &= aX - bXY \\ \dot{Y} &= cXY - mY\end{aligned}$$

Find the conservation law hiding in this model.

*Solution*

Using the approach from the chapter we can describe trajectories by

$$\frac{\mathrm{d}X}{\mathrm{d}Y} = \frac{\dot{X}}{\dot{Y}} = \frac{aX - bXY}{cXY - mY} \tag{1}$$

We can also write this as

$$\frac{\mathrm{d}X}{\mathrm{d}Y} = \frac{X(a - bY)}{Y(cX - m)}, \tag{2}$$

and hence

$$\frac{cX - m}{X}\mathrm{d}X = \frac{a - bY}{Y}\mathrm{d}Y \tag{3}$$

which we can integrate

$$\int \frac{cX - m}{X}\mathrm{d}X = \int \frac{a - bY}{Y}\mathrm{d}Y \tag{4}$$

to find

$$cX - m\ln(X) = a\ln(Y) - bY + K \tag{5}$$

where $K$ is the constant of integration. This yields the desired conservation law

$$K = cX + bY - m\ln(X) - a\ln(Y) \tag{6}$$

This conservation law describes a cycle in the state space on which the species oscillate.